MILITARY SPECIFICATION

COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS FOR CHARGE, PROPELLING, 155MM, M203A1

This Specification is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the component parts of the propelling charge designated as charge, propelling, 155MM, M203A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and Supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-70462 - Charge Propelling, 155MM, M203A1 Loading, Assembly and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing
MIL-STD-1168 -Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9345095 - Rear, Cap
9345097 - Igniter Container
9345098 - Body, Molded
9345109 - Composition
9357034 - Composition
9390469 - Packing and Marking for Box, Fiberboard for Igniter Container and Rear Cap, for Case Cartridge Assembly, 155MM M203A1
9390470 - Packing and Marking for Box, Fiberboard for Body, Molded for Case Cartridge Assembly, 155MM M203A1
9390475 - Packing and Marking for Box, Wood, Cleated Plywood for Box, Fiberboard for Igniter Container and Rear Cap for Case Cartridge Assembly, 155MM M203A1
9390476 - Packing and Marking for Box, Wood, Cleated Plywood for Box, Fiberboard for Body, Molded for Case Cartridge Assembly, 155MM M203A1
9345091 - Case Cartridge Assembly 155MM M203A1

(Copies of specifications, standards, drawings, publications and other Government documents required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D638 - Test Method for Tensile Properties of Plastics

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1. Materials. The constituent materials shall comply with the requirements of all applicable drawings and specifications.

3.2. Parts. The combustible cartridge case parts shall comply with all the requirements specified in applicable drawings and specifications.

3.3. Stability. The nitrocellulose in the combustible cartridge case parts shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested in accordance with 4.5.2.

3.4. Physical properties. The weights and tensile strength of combustible cartridge case parts shall meet the requirements of Table I.

TABLE I

Weight and tensile strength requirements for combustible cartridge case parts.

Part Description	Drawing	Weight	Tensile Strength	Density (1)
Rear Cap	9345095	2.0±0.25oz	N/A	0.85±0.12g/cc
Igniter Cont.	9345097	6.8±0.49oz	N/A	0.85±0.12g/cc
Body, Molded	9345098	29.0±1.64oz	4600±2000psi	0.98±0.12g/cc

Notes (1) Density is advisory

3.5. 50 - lb static test. The assembly of igniter container and body shall pass the 50 - lb static test and no separation of igniter container from the assembly shall occur when tested in accordance with 4.5.1.

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3.6. Proving ground (see 4.5.7). Ballistic tests shall be performed for First Article (see 4.3.2) and the first lot of each new interfix at 70°F, 145°F and -65°F. The cartridge case components, when assembled and fired as 155MM M203A1 propelling charge shall be consumed such that there is no inert or burning combustible cartridge case residue in the gun tube.

3.7. First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8. Workmanship. All parts shall be fabricated in a thorough, workmanlike manner, and shall be free of molding flaws, cracks, warpage outside the material envelope, crevassing, soft spots, frayed edges, foreign material greater than 1/4 inch in any dimension.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

Prior to dimensional examination of combustible cartridge case, components shall be conditioned for a minimum of twelve (12) hours at 23° Centigrade $\pm 2°$ and at 50% $\pm$5% relative humidity.

4.3. First article inspection.

4.3.1. Submission. The contractor shall submit a first article sample, as designated by the contracting officer for first article tests. The sample shall consist of parts which have been produced by the contract or furnished by a supplier which have been manufactured using the same production process, procedures, molds, dies and equipment which will be used in fulfilling the contract. If parts are produced from more than one (1) mold or die, they shall be identified by a suitable mold or die mark. Samples shall be taken equally from each mold or die used in production. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the

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requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, or source of supply as to significantly affect product uniformity as determined by the Government). The first article sample shall consist of the following items in sample quantities as indicated:

Part description	Drawings	Quantity
Body, Molded	9345098	80
Rear Cap	9345095	80
Igniter, container	9345097	80

In addition, whenever there is a lapse in production for a period in excess of 90 days, a first article sample, as directed by the contracting officer shall also be submitted as follows:

Part description	Drawings	Quantity
Body, Molded	9345098	50
Rear Cap	9345095	50
Igniter, container	9345097	50

4.3.2. Inspection to be performed. The first article parts consisting of quantity of 80 submitted in accordance with para 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table II. The first article parts consisting of quantity of 50 submitted in accordance with para. 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings or specifications and to the inspection of Table II with the exception of firing tests. All inspections listed in Table II shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3. Rejection. See MIL-A-48078

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Combustible Cartridge Case Components		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body, Molded (Drawing 9345098) Examination for Defects	50		3.2	4.4.2. 2.1
	Igniter Container (Drawing 9345097) Examination for Defects	50		3.2	4.4.2. 2.2
	Rear Cap (Drawing 9345095) Examination for defects	50		3.2	4.4.2. 2.3
	Combustible Cartridge Case Components (all) assembled as part of the M203A1 propelling charge, Firing Tests at 70°F, 145°F, and -65°F	30		3.2	4.5.7

NOTES:

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4.4. Quality conformance inspection.

4.4.1. Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and MIL-STD-1168. In addition, inspection lots of parts for case shall contain:

a. Nitrocellulose from one manufacturer and from not more than two lot numbers with the same interfix number.

b. Diphenylamine from one manufacturer and from not more than two lot numbers with the same interfix number.

c. Resin from one manufacturer and from not more than two lot numbers with the same interfix number.

d. Kraft from one manufacturer and not more than two lot numbers with the same interfix number.

e. Talc from one manufacturer and not more than two lot numbers with the same interfix number.

4.4.2. Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2.1. Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.2. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6.3. herein.

4.4.2.2. Classification of defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9345098
4.4.2.2.1	Body, Molded		NEXT HIGHER ASSEMBLY 9345103

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Material thickness		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
104	Straightness		0.40%	3.2	Gage
105	Weight		0.40%	3.4	4.5.6
106	Composition	4.4.3.3		3.2	4.5.3
107	Tensile Strength	4.4.3.4		3.4	4.5.4
108	Density	4.4.3.5		3.4	4.5.5
109	Presence of ribs within 2.00 in from rear end	1/		3.2	Gage
110	Diameter of holes (4 places)		0.40%	3.2	Gage
111	Unequally spaced holes (4 places)	1/		3.2	Gage
112	Nitrocellulose stability	4.4.3.2		3.3	4.5.2
Minor					
201	Outside radius	1/		3.2	Gage
202	Improper chamfer (angle and length)		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: 1/ Tool Controlled Dimensions- Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to production.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9345097
4.4.2.2.2	Igniter Container				NEXT HIGHER ASSEMBLY 9345103
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum outer diameter of igniter container		0.40%	3.2	Gage
102	Outer diameter of rear lip/rear cap interface		0.40%	3.2	Gage
103	Length of container		0.40%	3.2	Gage
104	Length of rear lip (0.60 in)		0.40%	3.2	Gage
105	Length of rear lip edge to rear body interface (2.8 in)		0.40%	3.2	Gage
106	Length of Maximum outer diameter		0.40%	3.2	Gage
107	Front outer diameter		0.40%	3.2	Gage
108	Thickness of material		0.40%	3.2	Gage
109	Weight		0.40%	3.4	4.5.6
110	Composition	4.4.3.3		3.2	4.5.3
111	Tensile strength	4.4.3.4		3.4	4.5.4
112	Density	4.4.3.5		3.4	4.5.5
113	Inside diameter of rear lip measured at intersection of 15° and rear surface of lip.	1/		3.2	Gage
114	Diameter of rear flat portion	1/		3.2	Gage
115	Length from rear edge of body interface to center of boss (4 places)	1/		3.2	Gage

NOTES: 1/ Tool Controlled Dimensions - Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to regular production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70461C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9345097
4.4.2.2.2	Continuation Sheet				NEXT HIGHER ASSEMBLY 9345103
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Diameter of bosses (4 places)	1/		3.2	Gage
117	Unequally spaced bosses (4 places)	1/		3.2	Gage
118	Length of flat surface of boss (4 places)	1/		3.2	Gage
119	Angle on boss (4 places)	1/		3.2	Gage
120	Nitrocellulose stability	4.4.3.2		3.3	4.5.2
Minor					
201	Improper chamfer (angle and length)		0.65%	3.2	Gage
202	Radius of rear outside corner	1/		3.2	Gage
203	Improper taper angle (7°)	1/		3.2	Gage
204	Improper taper angle (15°)	1/		3.2	Gage
205	Diameter of holes	1/		3.2	Gage
206	Number of holes	1/		3.2	Gage
207	Holes not properly spaced	1/		3.2	Gage
208	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345095
4.4.2.2.3	Rear, Cap				NEXT HIGHER ASSEMBLY 9345092
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of cap		0.40%	3.2	Gage
102	Inside diameter of cap		0.40%	3.2	Gage
103	Material thickness		0.40%	3.2	Gage
104	Length of Cap		0.40%	3.2	Gage
105	Weight		0.40%	3.4	4.5.6
106	Composition	4.4.3.3		3.2	4.5.3
107	Tensile strength	4.4.3.4		3.4	4.5.4
108	Density	4.4.3.5		3.4	4.5.5
109	Diameter of hole	1/		3.2	Gage
110	True position of hole	1/		3.2	Gage
111	Diameter of rear flat portion	1/		3.2	Gage
112	True position of flat portion	1/		3.2	Gage
113	Presence of ribs on inside	1/		3.2	Gage
114	Nitrocellulose stability	4.4.3.2		3.3	4.5.2
Minor					
201	Improper taper angle	1/		3.2	Gage
202	Improper chamfer of hole	1/		3.2	Gage
203	Improper chamfer of lip	1/		3.2	Gage
204	Improper inside radius	1/		3.2	Gage
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTE: 1/ Tool Controlled Dimensions - Whenever a new tool is introduced, 25 samples shall be checked for conformance with the dimension requirements for qualification of tool prior to regular production.

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4.4.3. Testing.

4.4.3.1. 50 - lb static test. Ten (10) molded bodies and ten (10) igniter containers shall be randomly selected from the first article sample and from each of the first three production lots and tested in accordance with 4.5.1. If any sample fails to pass the test, the lot shall be rejected. When three consecutive lots have been accepted, sampling may be reduced to five (5) molded bodies and five (5) igniter containers from each lot. If a failure occurs, sampling shall revert to ten (10) molded bodies and ten (10) igniter containers per lot until three consecutive lots have been accepted. Any failure shall be cause for lot rejection.

4.4.3.2. Stability. To determine the stability of the molded body, igniter container and rear cap, random samples of three (3) moldings shall be taken from each slurry batch making up a molded body, rear cap or igniter container lot. When three (3) consecutive lots of molded bodies, rear caps, or igniter containers have been accepted, sampling may be reduced to one (1) molding from each batch. In the event of a failure, sample size shall revert back to three (3) per batch. From each such sample, the stability shall be determined in accordance with 4.5.2. Any failure shall be cause for batch rejection.

4.4.3.3. Composition. To determine the composition of the molded body, igniter container and rear cap, one molding shall be withheld and marked immediately after pressing the raw felt from each slurry batch mixture or one molding from the beginning, middle and end of each batch production (when the slurry batch serves for more than one day's manufacture) of the lot. From each such sample, the composition shall be determined in accordance with 4.5.3. Failure of any specimen is cause for rejection of all parts manufactured from that batch since the last previously acceptable analysis and until an acceptable analysis is performed on a subsequently manufactured sample from that batch.

4.4.3.4. Tensile strength. Five (5) molded bodies shall be randomly selected from the first article and each production lot to provide the required test specimens for tensile tests in accordance with 4.5.4. Failure of any specimen is cause for lot rejection.

4.4.3.5. Density. Five (5) molded bodies, five (5) rear caps and five (5) igniter containers shall be randomly selected from the first article sample and each production lot to provide the test specimens for density tests in accordance with 4.5.5. Testing is for informational purpose only.

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4.5. Methods of inspection.

4.5.1. 50 - lb static test. The body shall be loaded with a 50 lb inert material and assembled together with the igniter container with the lock-in device. The inert loaded assembly shall be lifted above the ground with the igniter container facing downward and clamped on a suitable test fixture for 5 minutes. Observation shall be made for separation of the igniter container from the body. Any separation occurrence in the testings would be cause for lot rejection.

4.5.2. Stability. The sample quantity of combustible case components shall be subjected to the stability heat test at 134.5°C in accordance with MIL-STD-286, Method 404.1.2.

4.5.3. Composition.

4.5.3.1. Preparation of specimen (see 4.4.3.3.). Each specimen shall be divided into 1/8-inch (approximate) cubes or equivalent. The specimens shall then be thoroughly mixed to form a composite. From the composite, approximately 1.0 g. shall be selected as the analysis sample and weighed to the nearest 0.2 milligram (mg), (weight A) in an extraction thimble. The same thimble should be carried through the extractions in 4.5.3.3.1. and 4.5.3.3.3.

4.5.3.2. Total volatiles. The samples from 4.5.3.1 shall be dried for 3 hours at 100±2°C, cooled in a desiccator (1/2 hr. min.), or placed in a microwave moisture analyzer and dry to constant weight and weighed to the nearest 0.2 mg (Weight W). Calculate the percent total volatiles as follows:

$$\%\text{Total Volatiles} = \frac{A - W \times 100}{A}$$

4.5.3.3. Analytical procedure.

4.5.3.3.1. Extraction. Using a tared, extra-coarse, sintered glass thimble, extract each sample from 4.5.3.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286, Method 104.1). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.3.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 20°C. Cool in a desiccator (1/2 hour min). Determine the dry weight of this residue (weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.3.3.3.

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4.5.3.3.2. Diphenylamine (DPA).

4.5.3.3.2.1 Method 1. To the dried extract from 4.5.3.3.1, add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with paragraph 5.5) of Standard MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodine solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in Standard MIL-STD-286, Method 201.4, paragraph 5.11.

4.5.3.3.2.2 Tentative method 2. (Informational test for the first 3 production lots) see 6.7. Prepare 1.2 to 1.5 grams of finely cut sample. Dry in a microwave oven. Transfer the dried sample to a weighing bottle. Cool the weighing bottle in a desiccator for 5 minutes and weigh. Transfer the sample to a square beaker. Reweigh the empty weighing bottle. Sample weight = total weight minus weight of empty weighing bottle. Add 50 ml of 70% methanol to the square beaker, grind the mixture by the use of the polytron homogenizer for 3 minutes or until the sample is completely defiberized. Transfer the mixture to a millipore filtering apparatus with 0.45 micron cellulose nitrate membrane. Use a squeeze bottle with 70% methanol to wash the square beaker. Add the washing solution on top of the residue. Add 50 ml of fresh 70% methanol to the square beaker. Grind this solution in the polytron for 1 minute. Use spatula to transfer the residue from the filtering apparatus to the square beaker containing the methanol. Place the beaker in an ultrasonic cleaner which has warm water (minimum 45°C) for 10 minutes. Transfer the mixture to the filtering apparatus without changing the filter. Use a squeeze bottle with 70% methanol to wash the square beaker and add the washing solution to the residue in the filtering apparatus. Repeat steps above two times except the time in the ultrasonic cleaner is reduced to 5 minutes. Transfer the filtrate into a 500 ml volumetric flask. Make up to the mark with 70% methanol. Shake the flask well enough to make sure the solution is uniform. Transfer exactly 15 ml of this solution into a 50 ml volumetric flask. Make up to mark with 70% methanol. Shake well. Measure the UV absorption of this solution at 285 nm. Calculate the DPA content according to the following formula:

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$$\text{DPA \%} = \frac{(20A - 0.2) \times 2.5}{32.001 \times W}$$

Where A = absorbance

W = sample weight in gms.

4.5.3.3.3. Nitrocellulose. To provide good heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.5.3.3.1 and extract with acetone using the method in 4.5.3.3.1. After extraction dry the residue remaining at 100± 2°C for one hour. Cool in a desiccator (1/2 hour min) weigh to the nearest 0.2 mg (Weight D). Evaporate the acetone extract to dryness and the inert atmosphere throughout this test. Reflux the solution for five (5) minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous amonium sulfate (prepared per MIL-STD-286, Method 604.1) and 10 ml concentrated hydrochloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammoniumthiocyanate solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate the percent NC as follows

$$\text{\% NC} = \frac{46.69N}{E} \times \frac{(V-B)}{W} \times 100$$

Where:

V = Volume titanous chloride used for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.3.3.4. Talc content (ash test). The specimen of approximately 1 - 1.5 grams shall be taken from the same case body for the above testing and weighed to within 0.2 mg. in a tared porcelain crucible. Cover the crucible. The specimen shall then be ignited at 500°C in a muffle furnace. The crucible shall be cooled to room temperature and then weighed. Heat until all evidence of carbon is gone (no black residue), or for one hour. The increase in weight of the crucible shall be calculated on a total volatile - free basis to the percent of talc content.

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4.5.3.3.5. Kraft, resin and additives. Calculate the percent kraft, resin and additives as follows:

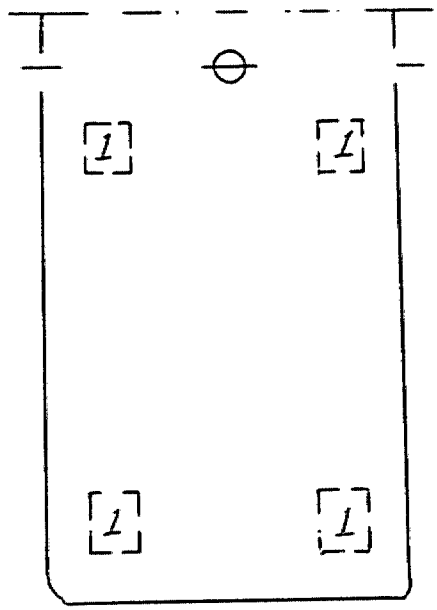
$$\% \text{ Kraft} + \% \text{ resin} + \% \text{ additives} = 100 - (\% \text{ NC} + \% \text{DPA} + \% \text{ Talc})$$

4.5.3.3.6. Report the total methylene chloride extractables $\frac{(w - C) \times 100}{W}$ and the total acetone extractables $\frac{(C - D) \times 100}{W}$

4.5.4. Tensile strength. Two specimens shall be prepared from each molded body. The specimens shall be taken from the center of the wall of the body. Test specimens shall be taken 180° apart. The specimens shall have a one-inch gage length and width, but wall thickness shall fall within the tolerance specified on the applicable drawing. The test specimens shall be preconditioned for a minimum of 4 hours at 23 plus or minum ($\pm$) 2° centigrade (C) at a relative humidity of 50 $\pm$ 5 percent (%). The tensile breaking strength of the specimen shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute.

4.5.5. Density. Six (6) specimens shall be taken from each molded body in accordance with Figure 1. Three (3) specimens shall be taken from each rear cap and igniter container sample in accordance with Figure 2 and Figure 3 respectively. All the specimens shall be dried for 3 hours at 105°C or for 12 hours at 70°C. The specimen thickness shall be measured, the volume computed and the sample weighed on a calibrated scale having an accuracy of $\pm$ 0.02g. The average density for each part shall be calculated and shall be in accordance with the applicable requirements.

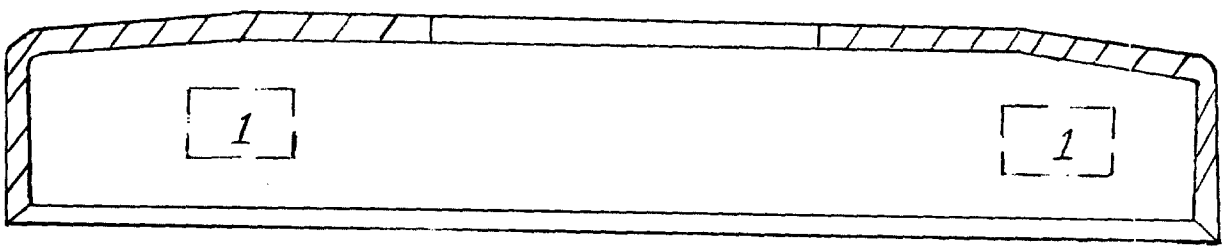
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1. Three samples, 1 in. X 3 in. displaced by 120° for testing the density.
(Each from top and bottom)

FIGURE 1. Body

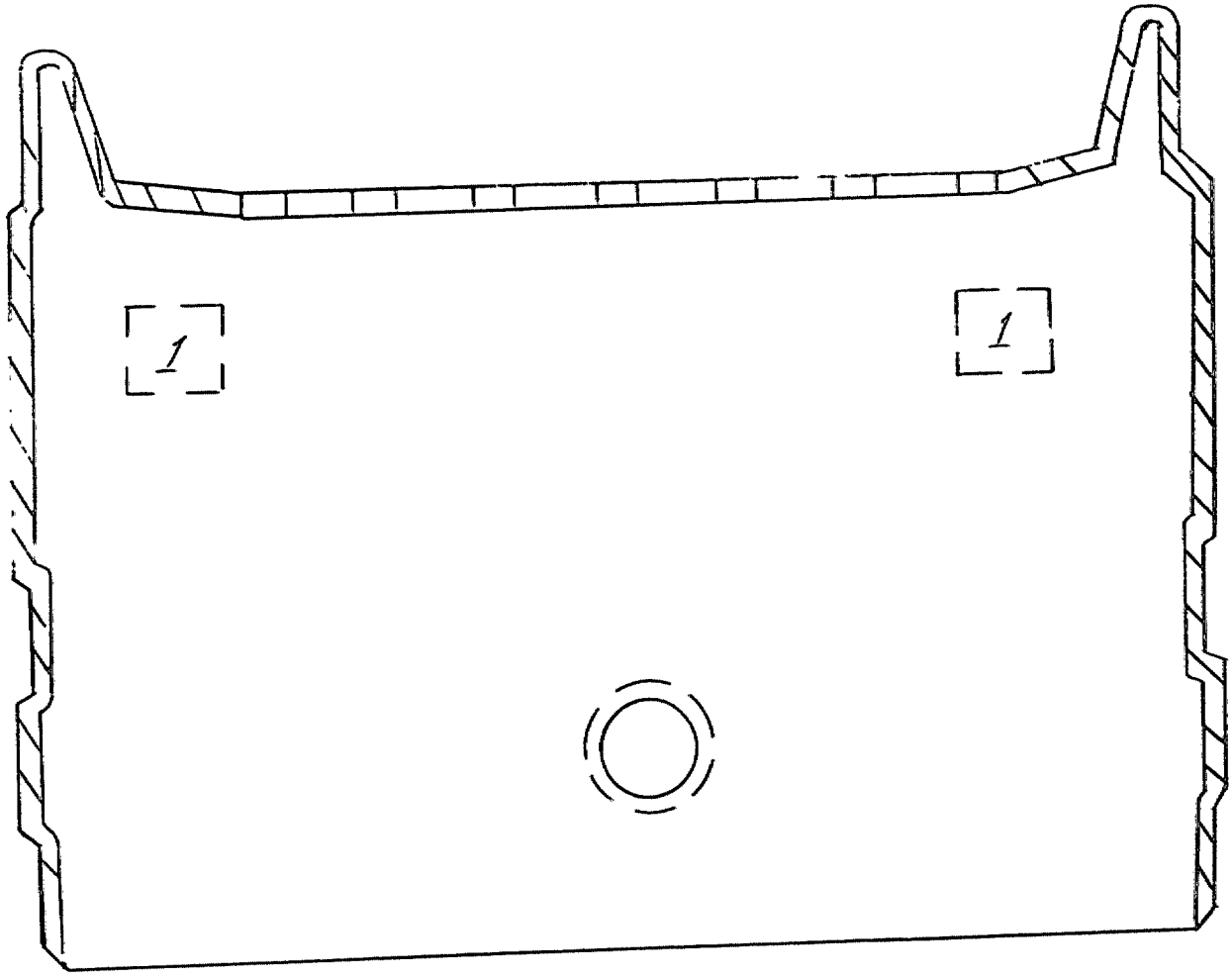
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1. Three samples, 1/2 in. by 1/2 in. displaced by 120° for density test.

FIGURE 2 Rear Cap

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1. Three samples, 1 in. X 1 in. displaced by 120° for density test.

FIGURE 3. Igniter container

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4.5.6. Weight. All samples shall be weighed individually, using any government approved and suitably calibrated scale capable of reading to the nearest 0.1 g. Test samples shall be preconditioned for a minimum of 4 hours at 23 ± 2 degrees C and 50 ±5% R.H. prior to weighing.

4.5.7. Proving ground firing. The combustible case components shall be assembled as 155mm M203A1 propelling charge with the established service charge load in accordance with MIL-C-70462. The tests shall be conducted at a Government proving ground in accordance with the applicable acceptance test procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired in cannon M199 with 75 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65°F to -60°F; 145°F from 140°F to 145°F; 70°F from 68°F to 72°F. The test samples shall be fired at the temperatures indicated in Table III and any inert or burning residue shall be cause for lot rejection. Camera coverage of the weapon chamber and tube shall be provided for all test firings.

TABLE III

Proving ground firing

Temperature	First Article	First Lot or Each New Interfix
-65°F	10	10
+70°F	10	10
+145°F	10	10

4.5.7.1. Refiring. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the results shall be declared invalid, and a new test shall be performed.

5. PACKAGING

5.1 Packing.

5.1.1 Packing for igniter container and rear cap.

5.1.1.1 Level A. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390475-1.

5.1.1.2 Level B. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390475-2.

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5.1.1.3 Level C. The Igniter Container and Rear Cap shall be packed in accordance with drawing 9390469.

5.1.2 Packing for body, molded.

5.1.2.1 Level A. The Molded Body shall be packed in accordance with drawing 9390476-1.

5.1.2.2 Level B. The Molded Body shall be packed in accordance with drawing 9390476-2

5.1.2.3 Level C. The Molded Body shall be packed in accordance with drawing 9390470.

5.2 Marking.

5.2.1 Marking for igniter container and rear cap.

5.2.1.1 Level A and B. Mark in accordance with drawing 9390475.

5.2.1.2 Level C. Mark in accordance with drawing 9390469.

5.2.2 Marking for body, molded.

5.2.2.1 Level A and B. Mark in accordance with drawing 9390476.

5.2.2.2 Level C. Mark in accordance with drawing 9390470.

6. NOTES

6.1. Intended use. The items covered by this specification is intended for use in M203A1 155MM Propelling Charge.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See 6.2.3. of MIL-A-48078. Submit equipment designs, as required to Commander, ARDC, ATTN: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

6.4. Submission of ammunition data cards. In addition to the normal distribution of records, when the cases are procured by the Army, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
ARDC
ATTN: AMSMC-QAR-R (D)
SMCAR-AEE-BP
Dover, New Jersey 07801-5001

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6.5. Excessive residue. Residue is defined as excessive if it is significant enough to prevent chambering or hinder functioning of a subsequent round, or exceed 1 inch by 2 inches in size, as determined by the proving ground.

6.6. Proving ground.

6.6.1. Cleaning. Prior to each firing, the gun breech shall be thoroughly cleaned and dried.

6.6.2. Handling and storage. Special care shall be taken in the handling and storage of sample cases to prevent contamination by grease, grime, dirt, or other foreign material.

6.7. Informational DPA content determinations. The tentative Method 2 for DPA content determinations is for informational purpose and cannot be used for acceptance test purpose. The informational test will be done for the first 3 production lots in addition to the normal acceptance test Method 1. All the above batch composition test results shall be submitted to Commander, ARDC, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801-5001 for statistical evaluation of equivalency of both methods. If the tentative Method 2 is found to be equivalent to Method 1, an ECP will be submitted for changing the tentative method to alternate method in this specification.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.9 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A885)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70461C	COMBUSTIBLE CARTRIDGE CASE ASSEMBLY PARTS FOR CHARGE, PROPELLING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
155MM, N203A1	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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